

狹義相對論

21-1 牛頓的絕對時空觀

伽利略變換：

$$\left. \begin{aligned} x' &= x - vt \\ y' &= y \\ z' &= z \\ t' &= t \end{aligned} \right\}$$

把方程組對時間 t 求導：

$$\left. \begin{aligned} u'_x &= u_x - v \\ u'_y &= u_y \\ u'_z &= u_z \end{aligned} \right\}$$

將上式合併成矢量式

$$\Rightarrow u' = u + v$$

把矢量式對時間 t 求導：

$$\frac{du'}{dt'} = \frac{du}{dt}$$

洛倫茲變換 $\left\{ \begin{array}{l} \text{相對性原理} \\ \text{光速在真空不變} \end{array} \right.$

設 k 為某個系數(常數)，則有

$$\left\{ \begin{aligned} x &= k(x' + vt') & \dots\dots\dots(1) \\ x' &= k(x - vt) & \dots\dots\dots(2) \end{aligned} \right.$$

協變性(1) $\{t' = t$

$$\text{光速不變(2)} \left\{ \begin{aligned} x' &= ct' \\ x &= ct \end{aligned} \right.$$

(1) $\times$ (2) :

$$xx' = k^2(x' + vt')(x - vt)$$

$$(ct)(ct') = k^2(ct' + vt')(ct - vt)$$

$$c^2 tt' = k^2(c + v)(c - v)tt'$$

$$c^2 = k^2(c + v)(c - v)$$

$$k^2 = \frac{c^2}{c^2 - v^2}$$

$$k = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$S \rightarrow S'$$

代入方程組:

$$\left. \begin{aligned} x' &= \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \\ y' &= y \\ z' &= z \\ t' &= \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned} \right\} \quad (1)$$

$$S' \rightarrow S$$

代入方程組:

$$\left. \begin{aligned} x &= \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} \\ y &= y' \\ z &= z' \\ t &= \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned} \right\} \quad (2)$$

相對論下的速度:

$$u_x = \frac{dx}{dt}, \quad u_y = \frac{dy}{dt}, \quad u_z = \frac{dz}{dt}$$

$$u'_x = \frac{dx'}{dt'}, \quad u'_y = \frac{dy'}{dt'}, \quad u'_z = \frac{dz'}{dt'}$$

對(2)兩端取全微分:

$$\left. \begin{aligned} dx' &= \frac{dx - v dt}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{u_x - v}{\sqrt{1 - \frac{v^2}{c^2}}} dt \\ dy' &= dy \\ dz' &= dz \\ dt' &= \frac{dt - \frac{v}{c^2} dx}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1 - \frac{v}{c^2} u_x}{\sqrt{1 - \frac{v^2}{c^2}}} dt \end{aligned} \right\} \quad (3)$$

所以我們有 $dx', dy', dz', dt' \xrightarrow{\text{去推}} u'_x, u'_y, u'_z$ 。

$$\left. \begin{aligned} u'_x &= \frac{dx'}{dt'} = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} \\ u'_y &= \frac{dy'}{dt'} = \frac{u_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{u_x v}{c^2}} \\ u'_z &= \frac{dz'}{dt'} = \frac{u_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{u_x v}{c^2}} \end{aligned} \right\} \quad (4)$$